



COMSATS University Islamabad, Lahore Campus

Department of Computer Engineering

Quiz 1 – FALL 2025

Course Title:	Digital System Design	Course Code:	CPE 344	Credit Hours:	4(3,
Course Instructor:	Dr. Muhammad Naeem Awais	Program Name:	BS Computer Engineering		
Semester:	7 th	Batch:	FA22	Section:	A
Date:	29 th September, 2025		Maximum Marks:	10	
Student's Name:			Registration Number:		

Question 1: [CLO1-PLO3-C5] [10 Marks]

Draw a state diagram and design a controller for a Door Unlocking System. The system consists of a Start button and two additional input buttons used to unlock the door. The system's transitions are defined as follows:

- The door unlocks only when the buttons are pressed in the correct sequence: 10 → 01 → 11, following the press of the Start button.
- If the sequence is not followed, the system generates an alarm and returns to its initial state.

